

Thomaz F S Bastiaanssen, PhD

*Bioinformatics, Microbiome-Gut-Brain Axis,
Theoretical Ecology, Medical Ecology*

Curriculum Vitae



✉ thomaz.bastiaanssen@ucc.ie
🌐 thomazbastiaanssen.github.io
🔍 Google Scholar - h-index: 36
ORCID
🐦 @thomazsan

Professional Summary

I am a bioinformatician interested in the interplay between the gut microbiome and host mood and mental health. The focus of my research currently lies in understanding microbiome-gut-brain communication from a theoretical ecology/bioinformatics perspective. In particular, I am interested in understanding the role of stability and volatility of the gut microbiome in anxiety and depression and in integrating different types of 'omics data in a biologically interpretable manner. As a bioinformatician, I value clear and easily interpretable analysis of complex data in order to promote interdisciplinary collaboration. I have a strong sense of justice and am committed to fostering a culture of inclusivity, equity and diversity in academia.

Employment

- 2024 - **Postdoc**, *Amsterdam University Medical Centres, VUmc*, Amsterdam
The role of host-microbial co-metabolism in depression and anxiety disorders
- Funded by NWO ZonMw Veni 2023 grant
- 2021 - 2024 **Postdoc**, *University College Cork*, Cork
Lead bioinformatician for the Cryanlab
- Awarded the '21/'22 APC Scientific Excellence award

Education

- 2018 - 2021 **PhD**, *University College Cork*, Cork
Thesis Title: Mining the Microbiome for Markers of Microbiota-Gut-Brain Communication and Mental Health
- Supervised by Prof. J.F. Cryan, Prof. T.G. Dinan and Dr. M.J. Claesson
- 2014 - 2017 **MSc**, *Utrecht University*, Utrecht
Research Trajectory: Molecular and Cellular Life Sciences - Bioinformatics
- 2010 - 2013 **BSc**, *Utrecht University*, Utrecht
Majored in Biology with a Minor in Art History

Research Contributions

As a bioinformatician with a background in molecular biology, microbiology and theoretical ecology, I have contributed to the microbiome field and, most significantly, the microbiome-gut-brain axis field.

- I have developed a method to measure microbial volatility and have associated it with severity of the host stress response. Volatility is now studied in many microbiome fields.
- My microbiome bioinformatics guidebooks have been used by several lecturers and research groups across many universities, both for research and educational purposes.
- I have developed and continue to maintain several free and open source bioinformatics and statistical analysis tools in the R programming language.

Mentorship

- 2022 - **Benjamín Valderrama**, *PhD candidate, University College Cork, Ireland*
- 2022 - **Emma Todd**, *PhD candidate, Deakin University, Australia*
- 2021 - 2022 **Alessandro Atzeni**, *PhD candidate, University of Rovira i Virgili, Spain*
- 2021 - 2022 **Samuele Laudani**, *PhD candidate, University of Catania, Italy*
- 2019 - 2020 **Simon Spichak**, *MSc candidate, University College Cork, Ireland*

Research Skills

Academic: Study development • Mentoring • Grantsmanship • Scientific writing

Bioinformatics: 16S analysis • metagenomic shotgun analysis • RNAseq analysis • Metabolomics analysis • Microbiome functional inference • Volatility analysis • Multi-omics integration

General Programming: R • tidyverse • R package development • Python • Bash • Server management

Statistics: Experimental design • Data analysis • Generalised linear models • Generalised linear mixed effects models • Principal component analysis • Compositional data analysis • Mediation analysis

Selected Grants, Honours & Awards

Grants: Principal applicant

- 2024 **The role of microbiome-immuno-metabolic interactions in depression and anxiety disorders**, *The Netherlands Organization for Scientific Research (NWO, ZonMw)*, Veni
Awarded: 320.000 €.

Grants: Co-applicant

- 2024 **Guts to beat depression!**, *The Netherlands Medical Research (ZonMw)*, Open Competition
Awarded: 878,129 €.
- 2024 **POWER**, *Horizon Europe, ERA4health, NutriBrain 2024*
Awarded: 331,908 €.
- 2023 **Role of prenatal stress in the co-development of children's gut microbiome and socioemotional problems**, *Canadian Institutes of Health Research (CIHR)*
Awarded: 1,735,000 CAD.

Honours

- 2021 - 2022 **APC Microbiome Ireland Scientific Excellence award**

Awards

- 2023 **European Society of Neurogastroenterology and Motility (ESNM)**, *Best abstract for Translational Science award*, Awarded during the NeuroGastro 2023 meeting in Bucharest.
- 2018 **4th Intl Symposium on Resilience Research**, *Best poster award*, Awarded during Resilienz meeting in Mainz.
- 2016 **GENIEUR travel grant**, To attend "Molecular Methods in IBS Research: genetics, epigenetics and microbiota research" in Heidelberg.
- 2015 - 2016 **ERASMUS+ Traineeship**, For MSc internship in Cork.

Board & Committee membership

Editorial Boards

- 2023 - **Gut Pathogens**, *Associate Editor*, Springer Journal

Scientific committees

- 2024 - **ECNP Early Career Academy**, *Inaugural member*
- 2024 - **ECNP Depression Meta-Network**, *Member*
- 2023 - **ECNP Resilience Network**, *Junior shell member*

Academic committees

- 2018 - 2023 **APC Wellbeing Committee**, *Founding member*, served as chair in 2021.

Organisational experience

- 2021 **Canadian Biomarker Integration Network for Depression (CAN-BIND)**, *Chaired session*.